

Blueprint for Healthy & Precision Longevity: Strategies for the Integration of Longevity Medicine into the National Health Service

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Abstract

Italy is the world's secondoldest country and is projected to reach an agedependency ratio of 1:1 by 2050. The prevalence of multimorbidity, frailty and disability already exceeds 60 % in adults ≥ 75 years with majority of both patients and caregivers being women, yet the Italian National Health Service (Servizio Sanitario Nazionale, SSN) remains hospital-centred, fragmented and chronically underfunded. Longevity medicine is an emerging, preventive and personalized approach to healthcare that aims to extend the healthy human lifespan (healthspan) and improve the quality of life as individuals age. By merging insights from the biology of aging, multi-omic biomarkers, artificial intelligence, and geroprotective trials within a translational discipline, longevity medicine proposes a preventative, personalized, and data-driven healthcare approach. Rather than merely focusing on extending lifespan, it prioritizes strategies aimed at improving healthspan—the period of life spent in good health—and enhancing the overall quality of life as individuals age. While longevity medicine is still primarily a conceptual and methodological framework rather than a set of established clinical tools, its emphasis on early intervention, risk stratification, and precision prevention offers substantial potential to improve the sustainable management of age-related diseases and reduce healthcare costs over time. Clear communication about realistic outcomes and limitations of existing strategies is essential for managing expectations and foster trust in this emerging field.

Despite its potential, practical applications of longevity medicine in Italy are still largely absent, highlighting an urgent need for strategic integration within the national healthcare policy framework.

Here we analyse demographic and epidemiological trends, evaluate the structural limitations of the SSN, outline the conceptual and scientific foundations of longevity medicine, advocate for genuinely multidisciplinary research pipelines, and identify five priority areas for strategic investment: (i) clinically validated biomarkers of biological age; (ii) interoperable digital monitoring platforms; (iii) bayesian adaptive multimodal trials; (iv) explainable-AI riskstratification tools and (v) curricula of professional medical training should include understanding of the complex relationship between aging and the development of pathology and disability. Embedding these pillars in national policy could convert the demographic challenge into a catalyst for sustainable innovation in population health.

Demographic and epidemiological landscape

Italian population contraction is accelerating: births dropped to approximately 370,000 in 2024, representing a 2.6% decrease compared to 2023 and a 5.8% decrease compared to 2022, marking a record low. Of these newborns, approximately 180,000 were male and 190,000 were female, reflecting sex-specific demographic trends. Deaths exceeded 660,000, with sex-specific differences: approximately 340,000 males and 320,000 females, raising the proportion of adults aged ≥ 65 years to 24.1%—the highest in the EU (European Union) [1]. The Italian National Institute of Statistics [1] projects a median age of 51.5 years and 6.4 million residents ≥ 80 years by 2050 [1]. While age is consistently linked to health deterioration and loss of autonomy, large heterogeneity persists: some develop multimorbidity and disability, whereas others remain robust until late life. In other words, there is often a gap between chronological and biological ageing (Figure 1 and 2). Studies have identified contextual factors that accelerate biological-age trajectories and amplify the demand for longevity-driven solutions [2].

Epidemiology: sex-based prevalence and incidence; socio-economic and regional gradients. Life expectancy differs by up to 3.8 years between northern and southern Italy, largely owing to disparities in education, income and preventive care [1, 3]. This was further confirmed by the InCHIANTI and BLSA cohorts, which showed that low socio-economic status accelerated epigenetic age and increased the incidence of multimorbidity [4-6].

Lifestyle. In Italy, population surveys indicate that adherence to the Mediterranean diet is frequently only moderate, with high adherence observed in less than 5% of adults [7]. Physical inactivity is highly prevalent among older adults, with surveillance data indicating almost half of Italians aged ≥ 65 are sedentary, contributing to increased obesity and cardiometabolic burden [7, 8].

Environmental toxic exposures. The majority of Italians are chronically exposed to fine particulate air pollution (PM_{2.5}, particles smaller than 2.5 micrometers) at concentrations exceeding the WHO safety threshold of 5 $\mu\text{g}/\text{m}^3$, posing a significant public health concern [9].

Exposome and life-course trajectories of ageing
The exposome - the cumulative set of physical, chemical, social, and behavioural exposures across the life course - drives heterogeneity in biological ageing and functional decline beyond the modest fraction explained by genetics. Critical windows (prenatal, early childhood, mid-life) link risk and protective factors with inflammatory and epigenetic signatures associated with multimorbidity and disability in later life [9-11]. In Italy, regional patterns of Mediterranean dietary adherence, social networks, and urban form, represent actionable levers for prevention [12, 13]. Integrating the exposome into precision longevity enables identification of high-risk subgroups and supports territory-specific policies and primary prevention within the National Health Service [14].

Beyond biomarkers: integrating multi-domain data.
Longitudinal assessment of biological age is strengthened by multi-omic cohorts that also capture socio-economic, environmental and cultural factors, improving the calibration of biological-age clocks and risk algorithms and quantifying contextual influences such as early-life deprivation, pollution and urban form [15-17]. In a country with marked regional variation, analytic pipelines that integrate clinical, omic, lifestyle and place-based data should support region-specific interventions and be evaluated not only on statistical performance but on decision utility, such as prevented events and improved functional trajectories [16-18].

Early-Life Intervention and Social integration as a longevity intervention. Social integration across the life course is a modifiable determinant of healthy longevity. Experiences in childhood, adolescence, and midlife shape later-life trajectories of physical, cognitive, and emotional function: early isolation or exclusion increases the risk of frailty and premature mortality, whereas stable, supportive relationships

build reserve and resilience [19–21]. This evidence implies that social integration should be targeted early, through schools, youth programmes, and workplaces that foster inclusion, prosocial skills, and civic participation, and then reinforced in midlife as social roles change [19–21]. In later life, when behaviours and networks are harder to modify, primary care and community services should prioritise older adults at high risk of loneliness and offer simple, regular opportunities for meaningful contact, including intergenerational activities and culturally relevant group programmes [22–25]. These interventions are intrinsic to geriatric care and exemplify a life-course philosophy of healthy longevity.

Biological-age heterogeneity among individuals highlights that chronological age alone is insufficient to capture the complexity of ageing processes. Deep-learning-based multi-omics clocks have been shown to predict mortality independently of chronological age, emphasizing the potential of biological-age markers in personalized healthcare. Composite biomarkers, such as the Pace of Ageing, can explain a substantial proportion of the variance in functional decline [16]. Additionally, biological assessments must account for sex-specific differences in ageing processes, as these can significantly influence health trajectories and therapeutic responses.

Multimorbidity, frailty and medication burden. In Italy, more than a quarter of community-dwelling older adults report polypharmacy (≥ 5 drugs), with prevalence rising sharply among the very old [26]. Evidence from Italian and European cohorts shows that polypharmacy is associated with frailty progression and increased mortalities [27]. Recent Italian reviews further highlight sex-specific patterns in prescribing and adverse drug reactions, underscoring the complexity of safe medication management in older adults. In parallel, emerging BIOAGE inflammatory scores that combine markers (IL-6, TNF- α , and MCP-1) have been shown to predict accelerated disability trajectories and mortality independently of chronological age [28, 29], providing a mechanistic link between chronic inflammation, frailty, and pharmacological vulnerability.

Social isolation and early-life adversity. A meta-analysis of 3.4 million participants linked social isolation to a 29% excess mortality: census data show that 49 % of Italians ≥ 65 years live alone [19]. Social isolation is higher among caregivers—about 80% of whom are women—and is linked to an increase of dementia risk [30]. In addition, childhood poverty, post-traumatic stress disorder, and exposure to violence (more prevalent among women) accelerate the GrimAge epigenetic clock by up to 2 years and increase the risk of mid-life multimorbidity by 40% [31–33].

Sex differences in ageing are pronounced and have significant implications for healthcare delivery. National registry data indicate that frail women accumulate more severe functional deficits and exhibit distinct prescribing patterns compared to men [34]. Although women generally have longer lifespans, they experience poorer overall health and a higher incidence of dementia, depression, and anxiety. Additionally, women are more likely to use medications such as analgesics, hypnotics, and anxiolytics, which increase the risk of adverse drug events (ADEs) [35]. Severe drug–drug interactions occur more frequently among women, leading to higher rates of hospitalization and increased healthcare costs. Furthermore, women account for approximately 80% of both paid and unpaid caregiving roles, thereby amplifying their exposure to physical, emotional, and financial burdens associated with caregiving responsibilities. Addressing these disparities through targeted interventions and gender-sensitive healthcare policies could substantially improve women's health outcomes and reduce overall healthcare expenditures [34].

Digital divide. Evidence from remote areas of Italy highlights a substantial digital health gap among older adults, where limited e-health literacy constrains the adoption of telemonitoring platforms [36].

Collectively, these demographic, socio-economic, behavioural, environmental, molecular, and healthcare-interaction layers, together with critical lifestyle factors such as sedentary behaviour and inadequate nutrition, converge to accelerate functional decline and increase healthcare service demand. This underscores the imperative for a longevity-medicine framework capable of detecting—and

modulating—biological ageing before an irreparable loss of resilience occurs [37-38]. Addressing sedentary lifestyles and promoting optimal nutrition emerge as essential, modifiable targets for interventions aimed at slowing biological ageing and preserving healthspan. Given these considerations, longevity medicine should be conceived not merely as a discipline dedicated to the care of older adults, but as a life-course approach that also targets middle-aged populations and is complemented by preventive policies addressing the long-term impact of early-life exposures and behaviors on aging trajectories.

Structural and systemic constraints of the SSN

With a life expectancy of 82.7 years yet only 10.4 healthy lifeyears remaining at age 65, Italy ranks last among Western European countries for healthspan [39]. This widening lifespan–healthspan divide highlights a structural mismatch between the biology of ageing and a care model that is still predominantly reactive rather than P4: predictive, preventive, personalised and participatory [40].

Fragmented, hospitalcentric pathways. Twenty-one regional and autonomous health authorities have produced marked heterogeneity in funding, staffing and performance. Digital fragmentation persists, with incomplete interoperability of electronic health records across regions and care settings [41]. Home-care coverage for older adults remains low, with wide regional variability and often low care intensity—despite PNRR targets of 10% by 2026 [42]. While overall hospitalisation rates fell from 2012 to 2019, then rebounded in 2021–2023 without reaching pre-pandemic levels, the non-fatal burden of many NCDs has largely stagnated, amid persistent regional inequalities [43].

Preventionaverse financing. Diagnosis-related group (DRG) reimbursement mechanisms structurally prioritise procedures over preventive services. In Italy, the share of resources devoted to prevention has long remained below national targets and only slightly above the OECD average. Evidence from international modelling studies suggests that narrowing this prevention gap would yield substantial long-term health and economic benefits. [44].

Underresourced multidisciplinary teams. In Italy, geriatricians represent a relatively small proportion of the specialist workforce, and comprehensive training in evidence-based geriatric care—encompassing deprescribing, functional coaching, and data-driven decision making—remains limited. Although national specialization schools provide structured curricula in geriatrics, the modest number of trained professionals and the scarcity of widespread educational programmes constrain the capacity of the health system to fully integrate geriatric principles into routine practice [45]

Digital and participatory inertia. The minority of rural adults ≥ 60 years possess sufficient ehealth literacy to engage with telemonitoring platforms [46, 47]. Nevertheless, early pilots—ProPCC (integrated frailty care) and ProACT (multimorbidity platform)—reduced emergencydepartment visits by 37 % and sustained >70 % engagement over 12 months. National scaling is impeded by the absence of HL7FHIR standards and performancebased payments [41, 48, 49].

Collectively, these demographic, socio-economic, behavioural, environmental, molecular, and healthcare-interaction layers, together with critical lifestyle factors such as sedentary behaviour and inadequate nutrition, converge to accelerate functional decline and increase healthcare service utilisation. Addressing sedentary lifestyles and promoting optimal nutrition remain immediate and actionable strategies to mitigate biological ageing, while simultaneously pursuing rigorous validation, standardization, and clinical guideline development for emerging omic-based and AI-driven approaches [50].

Currently, the great part of healthcare spending is devoted to individuals who already live with chronic diseases and disabilities. Without a decisive shift toward primary prevention and longevityoriented care, even substantial improvements in traditional treatment pathways will only defer—and not avert—the accelerating accumulation of multimorbidity [40, 51]. This trajectory threatens to overload healthcare

systems in Italy and worldwide, underscoring the urgency of reallocating resources toward proactive strategies that preserve physiological resilience before chronic conditions take hold. Despite rapid progress in multi-omic, genetic and AI-based tools, they are still rarely used in everyday clinical practice. A recent multicentre trial improved continuity of care but did not change biological-age measures, suggesting that these precision tools must be applied earlier and linked to prevention. This gap, together with the structural weaknesses described above, reinforces the need to shift from mainly reactive treatment to proactive strategies that slow biological ageing and maintain resilience across the life course.

Conceptual foundations of longevity medicine

Longevity medicine is an evidencebased, geroscienceinformed framework that aims to quantify, understand and modulate the biological processes driving functional decline before overt disease manifests. Within the SSN it should build on—and not compete with—classical geriatrics, extending geriatric principles of personcentred, functionoriented care across the life course through several interlinked pillars.

1. **Longitudinal measurement of biological ageing.** Multi-omics ageing clocks, especially proteomic and DNA-methylation clocks, predict mortality, multimorbidity and physical function beyond chronological age, with recent proteomic clocks and epigenetic signatures also linked to gait-speed decline and other performance measures [52]. Continuous sensor-derived metrics (e.g. gait cadence, heart-rate variability) from consumer wearables can further refine these ageing trajectories at population scale [53]. Integrating longitudinal molecular and functional data in this way provides a powerful resource for longevity medicine, improving risk prediction and supporting the design of preventative therapies to extend healthspan.
2. **Mechanistic targets and lifestylefirst interventions.** The Hallmarks of Ageing are best viewed as a conceptual map of conserved molecular processes—none currently has a validated, clinically useful assay in humans, and only a handful can be quantified reproducibly in animal models. Even so, the pathways they outline intersect with lifestyle behaviours that are both actionable. Cohort analyses and randomised lifestyle trials consistently show that Mediterranean-style dietary patterns, adequate diet, regular aerobic and resistance exercise, restorative sleep, and structured stressreduction attenuate cardiometabolic risk, maintain physical function, and delay clinical onset of multimorbidity [54]. These benefits appear to operate through broad effects on nutrient sensing (AMPK, mTOR), systemic inflammation, mitochondrial efficiency, and epigenetic regulation—mechanisms that, while biologically plausible, remain proxy outcomes rather than direct measurements of a single measurable “hallmark” [50]. Pharmacological geroprotectors—including metformin, rapalogs, and emergent senolytics—are still under active investigation. Human evidence is limited to small proof-of-concept trials, and the field lacks robust biomarkers to confirm target engagement or predict long-term benefit [55, 56]. Experimental approaches such as heterochronic plasma exchange or partial cellular reprogramming are at an even earlier stage, confined to exploratory studies or animal models and burdened by safety and feasibility concerns [57]. As such, these agents should be regarded as prospective adjuncts—not replacements—for lifestyle interventions that already provide clear clinical benefit. Parallel efforts to develop reproducible, clinically interpretable biomarkers of biological ageing will be essential before any pharmacological strategy can be confidently deployed in routine practice. Indeed, developing measures of the biological mechanisms of aging is also fundamental for the next generation of Geroscience trials.
3. **Data-driven personalisation and participatory implementation.** Biological ageing measures such as DNA-methylation clocks and the DunedinPACE biomarker predict morbidity and functional decline beyond chronological age and are already being used as endpoints in European geroprotective trials, including studies of metformin [16,58].

Participatory digital platforms such as ProACT allow older adults with multimorbidity to engage with sensor-derived health data and support self-management [48]. Together, these tools show how integrating multi-omic biomarkers with patient-centred digital health can enable more personalised and adaptive prevention strategies in ageing medicine.

4. **Re-architecting health-system infrastructure for precision longevity.** To shift from reactive disease care to proactive healthspan optimisation, the SSN must be organised around longitudinal data integration, interoperable digital platforms and reimbursement models that reward prevention. Regional primary-care hubs should combine remote monitoring, AI-enabled risk-stratification dashboards and clear referral pathways to multidisciplinary longevity services, supported by secure national biobanks, privacy-preserving analytics and joint public–private innovation funds. Embedding these components in updated legislative frameworks would align incentives with the goal of extending healthy, productive years rather than merely treating late-stage disease.

Finally, realworld deployment hinges on regulatory recognition of healthspan extension as a valid endpoint; both AIFA and EMA will need to update guidance to accommodate composite functional outcomes and biomarkerdriven trial eligibility.

The case for multidisciplinary translational research

Delivering the promises of geroscience requires convergence across basic biology, clinical medicine, trial methodology, data science, social research and health policy. Geroscience trials face distinctive challenges in defining target populations, selecting meaningful outcomes and developing reliable biomarkers, so stronger integration across these domains is essential, even though evidence on the impact of interdisciplinarity remains limited [59].

Living laboratories and distributed cohorts. The NIH Translational Geroscience Network, European biobank infrastructures such as BBMRI-ERIC and the Italian Age-It programme provide harmonised protocols and large, distributed cohorts that can accelerate early-phase geroscience trials [60–62].

Regulatory and health-economic interfaces. Early dialogue with regulators (AIFA, EMA) can speed acceptance of novel, biomarker-based endpoints and adaptive designs, while embedding health economists in trial design ensures early evidence on costs and QALYs to support scalability, reimbursement and policy uptake [60,63,64].

Behavioural and implementation science. Simple digital nudges (e.g. SMS reminders), brief peer coaching and integrated platforms such as ProACT improve adherence to preventive interventions and support scalable, real-world implementation for older adults with multimorbidity [65,66].

Establishing an Italian “Gero-Hub”—modelled on comprehensive cancer centre networks—would consolidate molecular, clinical and data resources, operationalise the strategic priorities outlined below and prime the SSN for biomarker-guided prevention at scale.

Strategic investment priorities

5.1 Biomarkers of biological age. Develop and validate biomarkers of biological age that can be used in routine care, including sex- and ethnicity-specific tools. Priority should go to standardising and externally validating integrated multi-omic clocks and engaging early with EMA to qualify biomarker-informed endpoints [67–69].

5.2 Interoperable digital monitoring platforms. Adopt national HL7 FHIR and open EHR standards so that data from wearables and home sensors can be integrated into SSN records, supporting scalable

telemonitoring that reduces hospitalisations, while inclusion vouchers and age-friendly design prevent a widening digital divide [41,47,70].

5.3 Bayesian adaptive multimodal trials. Use platform trials with shared control arms and Bayesian adaptive designs to test nutritional, pharmacological, physical and cognitive interventions in parallel, improving efficiency and accelerating translation of gerotherapeutics [71].

5.4 Explainable AI risk stratification. Deploy privacy-preserving, explainable AI models (for example, federated learning) to predict frailty and other risks, with mandatory bias audits and documentation standards aligned with the EU AI Act and international best practice [72].

5.5 Age-proof workforce and citizen literacy. Introduce a “longevity curriculum” in medical and specialist training to build skills in multimorbidity and healthy ageing, and combine this with population-level communication strategies (such as mass media campaigns) to improve health literacy and participation in preventive services [73,74].

Together, these investments would establish the technical, digital, regulatory and human capital foundations required to embed longevity medicine within the SSN.

Outlook — from pilots to a national healthspan revolution

Italy's demographic trajectory is fixed, and the COVID-19 pandemic showed its frailty, yet its repercussions for the SSN are modifiable. The preceding sections demonstrate that scientific, digital and organisational assets are already in place; the remaining challenge is orchestration. Scalable longevity-enabling tools already exist and can ease this transition. Continuous glucose monitors, wearable activity and sleep sensors, AI-driven dietary coaching apps, and digitally delivered “exercise-as-medicine” programmes have each demonstrated real world gains in metabolic control, blood pressure reduction, fall prevention, and frailty delay. Deployed through primary care networks and supported by outcome-based reimbursement, these interventions can reduce the age-specific incidence and prevalence of chronic diseases and disability, slowing the accumulation of multimorbidity and lightening the societal and economic burden on the healthcare system.

Demonstration → diffusion. By 2027 the AGE-IT hub will supply biomarker-driven evidence at population scale. Embedding these tools in three ‘living-lab’ regions—selected for socio-demographic heterogeneity—would generate comparative data on cost-effectiveness, equity and acceptance, accelerating nationwide uptake.

Regulatory realignment. A joint AIFA–EMA fast-track for gerotherapeutics, and epigenetic age composites [75] could cut time-to-market by ~30 %. Aligning SSN reimbursement with *healthspan gains*—via value-based payments and adaptive tariffs—would create pullthrough for industry and clinicians alike.

Workforce and culture shift. Certifying 2000 longevity clinicians and deploying 10000 digital health navigators by 2030 would address the current geriatric and digital literacy gaps. Progress should be benchmarked against WHO Decade of Healthy Ageing indicators—functional ability, intrinsic capacity and caregiver burden [40].

National coordination. A dedicated Gero Hub—granted NCI-like authority—could pool public-private funding, enforce interoperability standards and publish an open dashboard of healthspan metrics, positioning Italy to spearhead a European Healthy Ageing Mission in line with the EU Green Paper on Ageing [76, 77].

Economic dividend. Economic evaluations indicate that intensive lifestyle interventions are consistently cost-effective for type 2 diabetes prevention, while metformin ranges from marginally cost-saving to cost-effective depending on the setting and time horizon. Microsimulation models at national scale

further suggest that large-scale prevention initiatives could generate substantial long-term savings—amounting to billions of euros or dollars—that could be reinvested into preventive services [78].

If these actions are legislated within the next parliamentary cycle, Italy can turn its status as one of the world's oldest nations into a living laboratory for equitable, sustainable healthy ageing, exporting evidencebased models across Europe.

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